1(a)	gravitational force (of attraction between satellite and planet)	**B1**
	<u>provides / is</u> centripetal force (on satellite about the planet)	**B1**
1(b)	$M = (4/3) \times \pi R^3 \rho$	**B1**
	$\omega = 2\pi / T$ **or** $v = 2\pi nR / T$	**B1**
	$GM / (nR)^2 = nR\omega^2$ **or** v^2 / nR	**M1**
	substitution clear to give $\rho = 3\pi n^3 / GT^2$	**A1**
1(c)	$n = (3.84 \times 10^5) / (6.38 \times 10^3) = 60.19$ or 60.2	**C1**
	$\rho = 3\pi \times 60.19^3 / [(6.67 \times 10^{-11}) \times (27.3 \times 24 \times 3600)^2]$	**C1**
	$\rho = 5.54 \times 10^3\ \text{kg m}^{-3}$	**A1**